

experiment Title :-

Study of LAN Topology, Network Devices,
and Address configuration.

Objective :- To study LAN Topology, identify networking devices, analyze wired and wireless communication measure network speed, and obtain system network.

Problem statement

1) Draw the network Topology:

.) Identify and draw the ~~can~~ LAN Topology of your college
(eg:- Star, Bus, Mesh)

.) clearly label all components, such as Router, switch
Access point, and end devices (PCs, laptops, etc)

Sol :- The college campus network examined during this study includes the following components:

.) layer-3 core switches installed in the server room for network management and routing.

.) POE (Power over Ethernet) switches used to supply power and connectivity to network devices.

.) wireless access points (APs) for providing wifi connectivity across the campus.

• Ethernet ports available in classrooms and for wired network access.

The college campus network uses a Hybrid Hierarchical topology that combines:

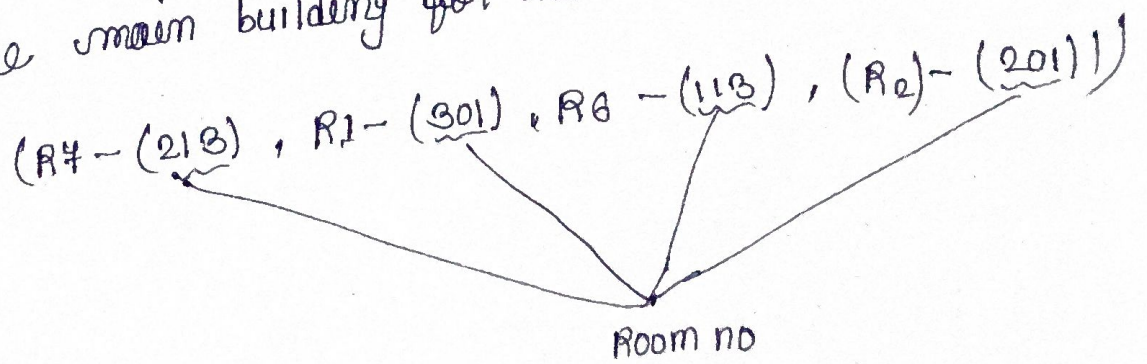
- Tree Topology
- Extended star Topology
- * BUS Topology
- * Ring Topology (2)

The network is divided into three layers:

- 1) Core layer
- 2) Distribution layer
- 3) Access layer

The core layer contains networking devices such as firewall router, layer-3 switches, patch rack and servers.

The distribution layer consists of four switches placed in different rooms/floors of the main building for network distribution.



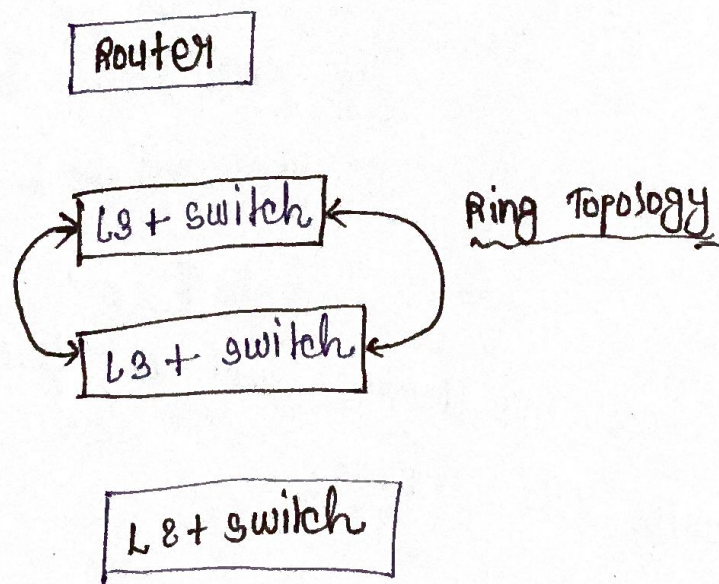
The Access layer contains three switches located in different rooms of the main building.

- R₁ - Room NO-301 (3rd floor)
- R₄ - Room NO-05 (ground floor)
- R₅ - Room NO-18 (ground floor)

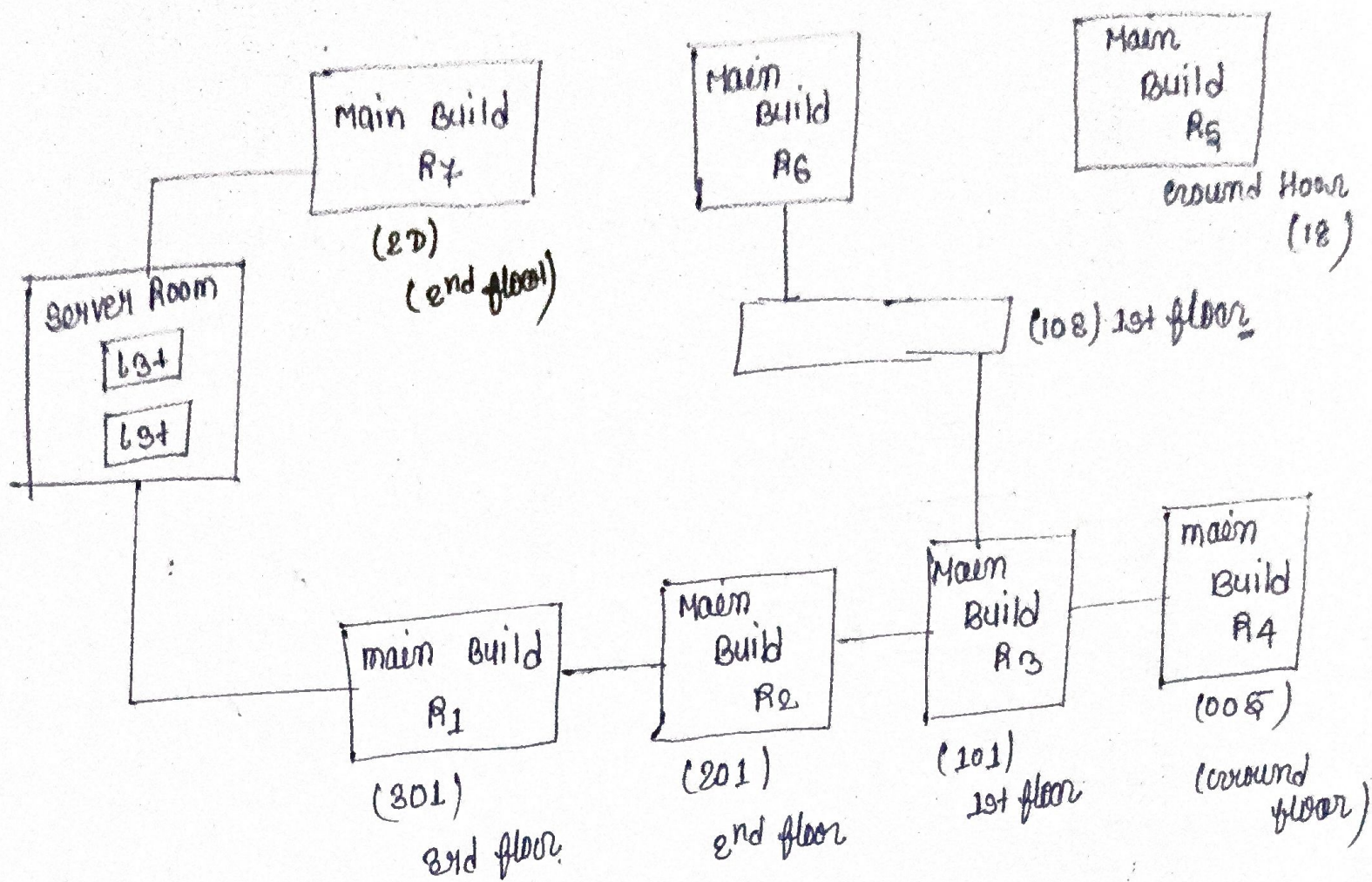
Access layer directly ~~connect~~ connect:

- ① Ethernet port
- ② wireless AP
- ③ Lab system
- ④ outdoor wireless AP (ex-018)

Server Room Architecture / Topology:



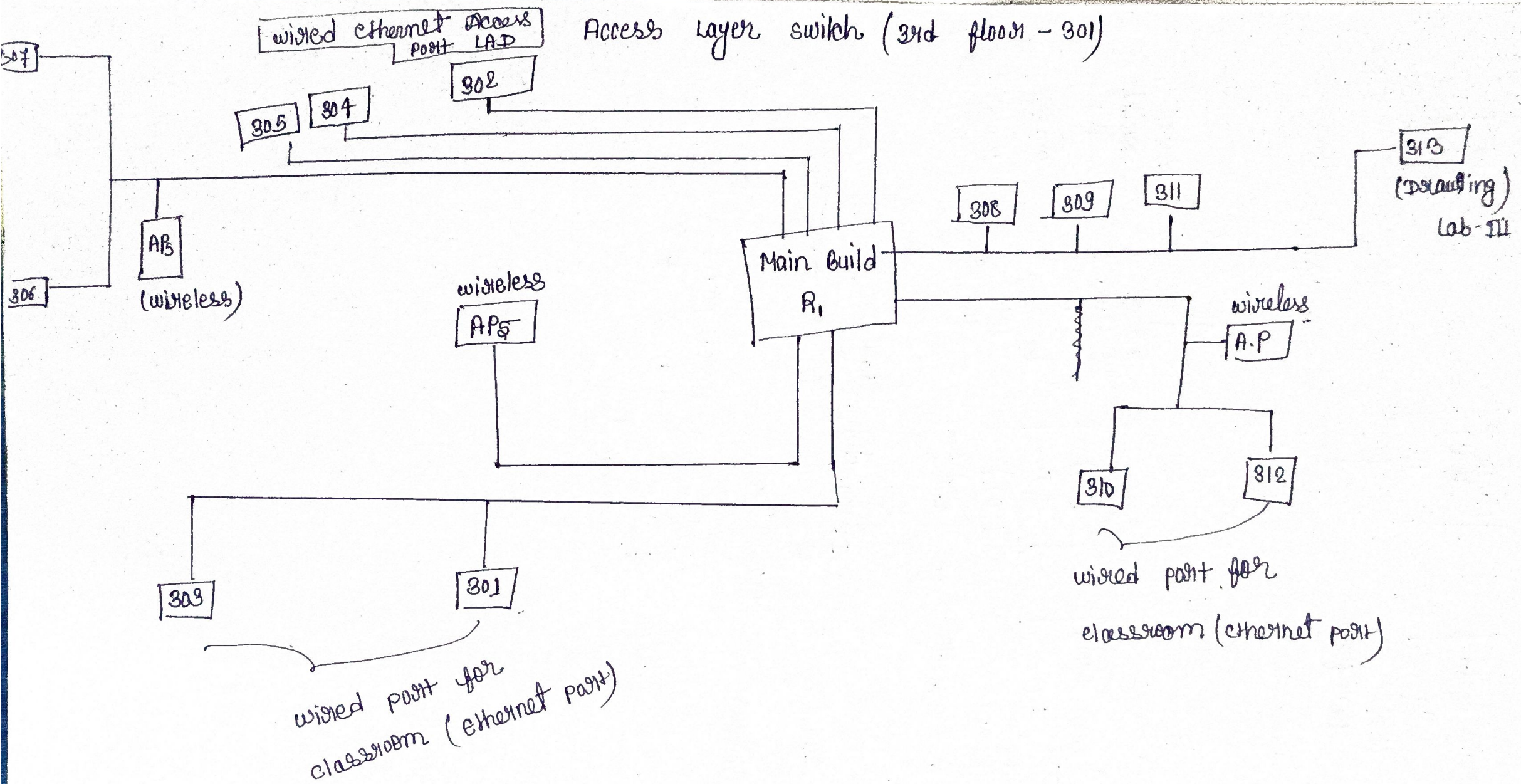
Topology



→ R₁, R₄, R₂, R₆ → Distribution layer

→ Server room → core layer

→ R₁, R₂, R₇, R₄, R₅, R₆, R₇ → Access point



3) Idan
 => The
 De
 Route
 Layer-3
 PoE switch
 wireless
 (AP)
 Firewall
 Solver

④ Identify network devices:-

.) List all networking devices used in the LAN

.) Mention the function of each device.

⇒ The following networking devices are used in the college LAN.

Device	function
Router	connects the local network to the internet and manages data routing.
layer-3 switch	perform switching and routing operations within the network.
POE switch	Provides both power and network connectivity to devices like Access point.
wireless Access point (AP)	Provides wi-fi connectivity to wireless users.
firewall	Protects the network from unauthorized access and cyber threats.
server	stores and manages network resources and services.

Ethernet cable

used for wired communication between devices.

End Devices

(Pcs/laptops)

used by users to access network resource and internet services.

③ Analyse connection types:-

The college campus networks use both wired (ethernet) and wireless (wifi) connections for communications.

* wired connections are used in :-

* computer laboratories

* server room

* desktop computers

* network switches and routers

→ ethernet cable are used to provide stable and high speed communications.

→ wireless connections (wi-fi) are used in :-

• laptops

• mobile phones

• smart phones

• classrooms and campus areas.

C:\Users\SAMRAT>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::c31b:2042:c3d8:211a%61
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Ethernet adapter Ethernet 4:

Connection-specific DNS Suffix . :
IPv6 Address. : 2401:4900:703c:cbb1:6af7:ba62:92d0:9880
Temporary IPv6 Address. : 2401:4900:703c:cbb1:4548:16c6:7202:846a
Link-local IPv6 Address : fe80::cebe:83f7:ff95:16e0%52
IPv4 Address. : 10.62.114.135
Subnet Mask : 255.255.255.0
Default Gateway : fe80::821:bfff:fed3:e72d%52
10.62.114.20

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet:

Media State : Media disconnected

Microsoft Windows [version 10.0.26200.8457]
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C:\Users\SAMRAT>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . . :
Link-local IPv6 Address : fe80::c31b:2042:c3d8:211a%61
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . :
IPv6 Address. : 2409:40e5:11f1:d271:a16b:6ef0:9df4:57f0
Temporary IPv6 Address. : 2409:40e5:11f1:d271:9010:cff4:12cd:3b4f
Link-local IPv6 Address : fe80::ae30:5c1d:812e:5221%18
IPv4 Address. : 10.28.95.74
Subnet Mask : 255.255.255.0
Default Gateway : fe80::485:60ff:fef4:a70e%18
10.28.95.69

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . . :

C:\Users\SAMRAT>

→ wi-fi Access points (APs) provide wireless internet connectivity across the campus.

④ Measure Network speed:

- find the link speed of

 - wired network (if available)

 - wireless network

- mention the tools/commands used (e.g - Task manager, netsh wlan show interfaces or etc).

Ans:- link speed is shown in the images in the next page at both wired and wireless.

The tools/command used to check the speed using:-

- Task Manager

- command prompt

command used

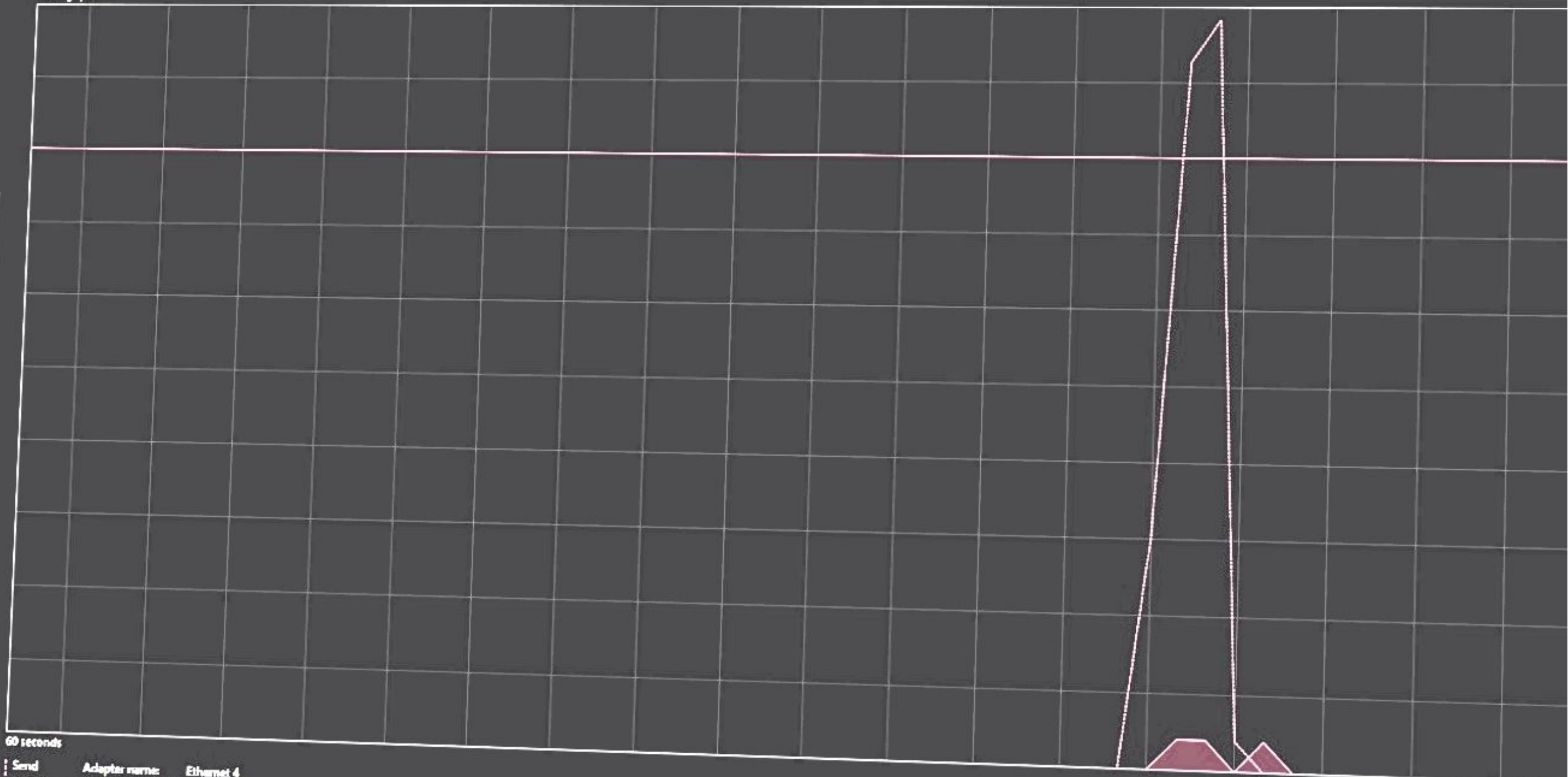
cmd

netsh wlan show interfaces.

Ethernet

Remote NDIS based Internet

Throughput



60 seconds

Send

0 Kbps

Receive

Adapter name:

Ethernet 4

Connection type:

Ethernet

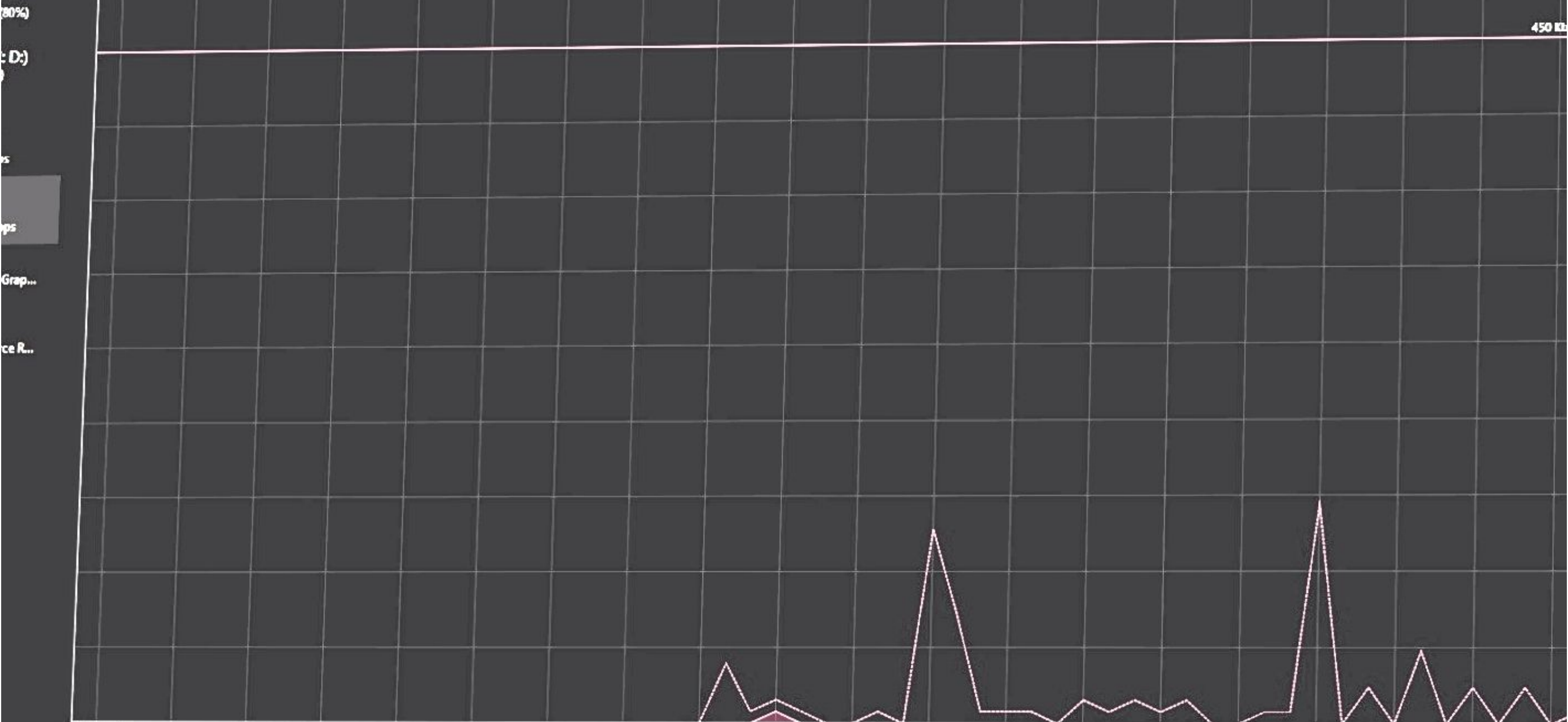
IPv4 address:

10.62.114.135

Wi-Fi

Intel(R) Wi-Fi 6E AX211 160MHz

Throughput



60 seconds

Send
8.0 Kbps
Receive
0 Kbps

Adapter name: Wi-Fi
SSID: moto g96 5G
Connection type: 802.11n
IPv4 address: 10.28.95.74
IPv6 address: 2409:40e5:11f1:d271::16b:6ef0:9df4:57f0
Signal strength: full

⑤ find system Network Details:

* Determine and record the following of your computers

— MAC addresses

— IP Address (IP V4)

— Subnet Mask

• Mention the command (li) used.

Ans: The record like Mac Address, IP Address (IPV4) and subnet Mask shown in next page.

The command is used :-

* To find IP Address and subnet Mask

→ ipconfig

* To find MAC Address

→ getmac

⑥ → difference between wired and wireless speed:-

⇒ There are various difference between wired and wireless speed:-

wired Network

- uses ethernet cable for communication
- generally provides higher speed
- More stable and reliable
- lower latency
- Better for gaming and large file transfer
- less affected by distance

wireless Network

- uses wi-fi radio signals
- speed may fluctuate
- Affected by signal interference.
- higher latency
- Better for mobility and convenience
- speed : decreases with distance

→ Role of subnet mask in networking:-

A subnet mask is used to divide an IP address into two parts:

* Network portion

* Host portion.

It helps devices identify whether another device is on the same network or a different network.

functions of subnet mask:-

- * identifies the network and host address.
- * controls network traffic
- * help in proper routing of data.
- * Improves network management and security.

examples:

IPv4 address: 192.168.56.1

subnet mask: ~~225.225~~ 255.255.255.0

~~in this example:~~

6

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conclusion

The experiment helped us in understanding LAN Topology, networking devices and network configuration. parameters, different connection types such as wired and wireless networks were studied. commands were used to identify Ip address, MAC address, Mac address. subnet mask, and network speed.

The experiment provided practical knowledge of computer fundamentals.